EX NO: 14 write a code using RAW sockets to implement packet sniffing

Aim

To write a program using RAW sockets to capture and display network packets passing through the network interface.

Algorithm:

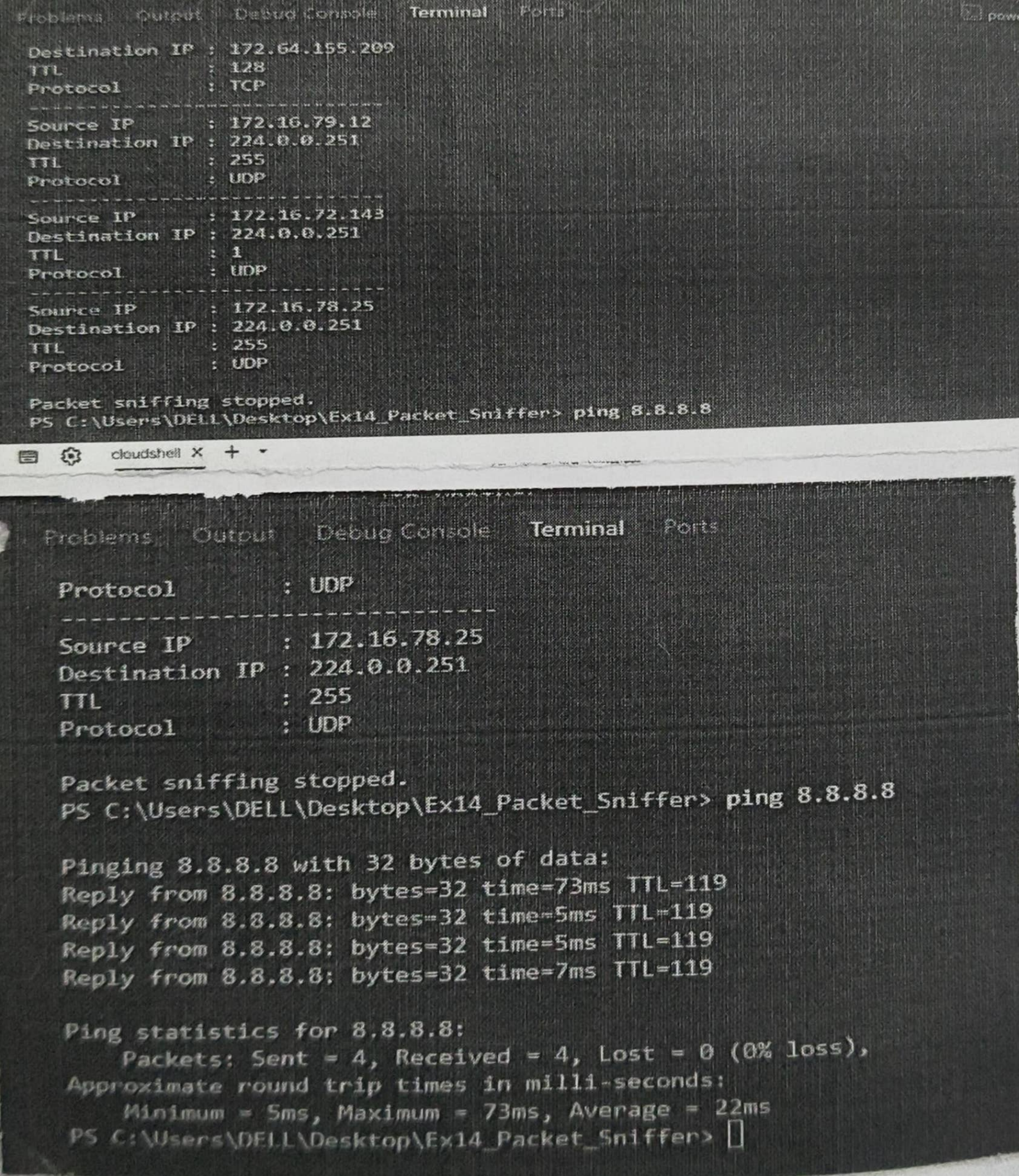
i) Start the program.
ii) Create an RAW socket using socket (AF_PACKET, SOCK_RAW, ...).
iii) Continuously receive packets from the network.
iv) Extract the Ethernet header from each packet.
v) Identify the protocol type such as IPv4, ARP or IPv6.
vi) For IPv4 packets, extract the source & destination IP addresses.
vii) Display the packet information.
viii) Repeat until the user terminates the program
ix) close the socket.

Program - Python

```
import socket
import struct
# Create a RAW socket
raw_socket = socket.socket ( socket.AF_PACKET,
            socket.SOCK_RAW, socket.ntohs(3))

print ("Packet sniffer started ...")
print ("Waiting for packets ...\n")

try:
    while True:
        # Receive packet
        packets, address = raw_socket.recvfrom (65535)
        # Ethernet header is first 14 bytes
        ethernet_header = packet [:14]
```



```
Problems  Output  Debug Console  Terminal  Ports

Destination IP : 172.64.155.209
TTL            : 128
Protocol       : TCP
--------------------------------
Source IP      : 172.16.79.12
Destination IP : 224.0.0.251
TTL            : 255
Protocol       : UDP
--------------------------------
Source IP      : 172.16.72.143
Destination IP : 224.0.0.251
TTL            : 1
Protocol       : UDP
--------------------------------
Source IP      : 172.16.78.25
Destination IP : 224.0.0.251
TTL            : 255
Protocol       : UDP

Packet sniffing stopped.
PS C:\Users\DELL\Desktop\Ex14_Packet_Sniffer> ping 8.8.8.8
```

```
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Problems  Output  Debug Console  Terminal  Ports

Protocol       : UDP
--------------------------------
Source IP      : 172.16.78.25
Destination IP : 224.0.0.251
TTL            : 255
Protocol       : UDP

Packet sniffing stopped.
PS C:\Users\DELL\Desktop\Ex14_Packet_Sniffer> ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=73ms TTL=119
Reply from 8.8.8.8: bytes=32 time=5ms TTL=119
Reply from 8.8.8.8: bytes=32 time=5ms TTL=119
Reply from 8.8.8.8: bytes=32 time=7ms TTL=119

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 73ms, Average = 22ms
PS C:\Users\DELL\Desktop\Ex14_Packet_Sniffer>
```

```
# Unpack Ethernet header
dest_mac, src_mac, protocol = struct.unpack(
        "! 6s6sH", ethernet_header)
dest_mac = ":".join("%.02x"%.byte for byte in dest_mac
)
src_mac = ":".join("%.02x"%.byte for byte in src_mac
)
print("Source MAC :", src_mac)
print("Destination MAC :", dest_mac)
print("protocol :", hex(protocol))
if protocol == 0x0800:
    ip_header = packet[14:34]
    version_ihl, tos, total_length, identification,
    flags_fragment, ttl, proto, checksum,
    src_ip, dest_ip = struct.unpack("!BBHHHBBH4s4s",
                        ip_header)
    src_ip = socket.inet_ntoa(src_ip)
    dest_ip = socket.inet_ntoa(dest_ip)
    print("Source IP :", src_ip)
    print("Destination IP :", dest_ip)
    print("TTL :", ttl)
    print("IP protocol :", proto)
elif protocol == 0x0806:
    print("Packet Type : ARP")
elif protocol == 0x86DD:
    print("Packet Type = IPv6")
else:
    print("Packet Type = other")
except keyboardInterrupt:
    print("\nPacket sniffing stopped.")
finally:
    raw_socket.close()
```

Result:

Thus, a packet-sniffing program using RAW sockets was successfully implemented. The program captures network packets and displays important information such as source MAC address, destination MAC address, protocol, source IP address, destination IP address & TTL.